Customers

Customer_Id	name
1	Rhofi
1	Mkeli

ORDERS

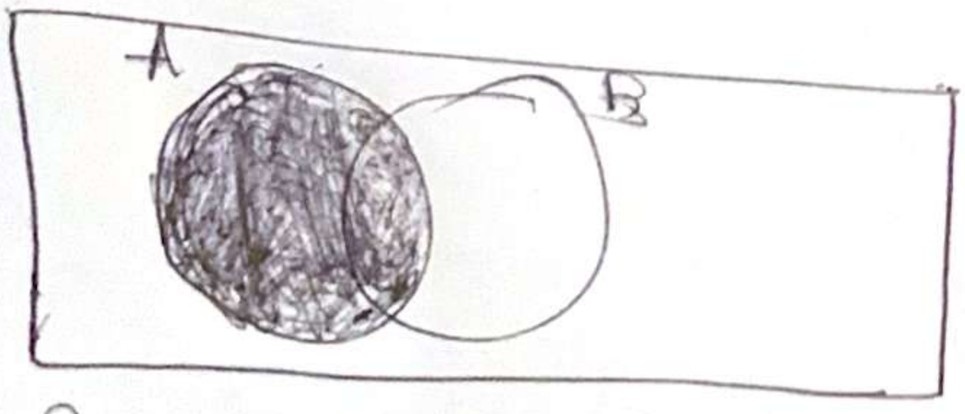
Order_Id	Customer_Id	amount
101	1	300
102	1	150
103	3	500

SYNTAX:
```
SELECT A.Customer_Id,
       A.name,
       B.amount
FROM Customers AS A
INNER JOIN Orders AS B
ON A.Customer_Id = B.Customer_Id;
```

Customer_Id	name	amount
1	Rhofi	300
1	Rhofi	150

LEFT JOIN

- Returns everything from the LEFT Table, and adds matches from the Right Table.
- If no matches, Right side becomes Null.



- Returns Everything on the left table.

```
SELECT Column(s) Specified
FROM Table1 AS A → LEFT table
LEFT JOIN table2 AS B → Right table
ON A.common_Column = B.common_column;
```

SQL JOINS

RELATIONAL Database → Structured Data.

101	102	103	104	105	106

Relationship

101	102	103	104	105	106

Primary Key - Is a unique identifier
- different in each Row
- Not Repeated.

Foreign Key -
- Can Be Repeated.
- Primary Key in another Table.

Customers primary key.

Customer-Id	Name
1	Rhofi
2	Nabali

ORDERS

Order-Id	Customer-Id	amount
101	1	300
102	1	150
103	3	500

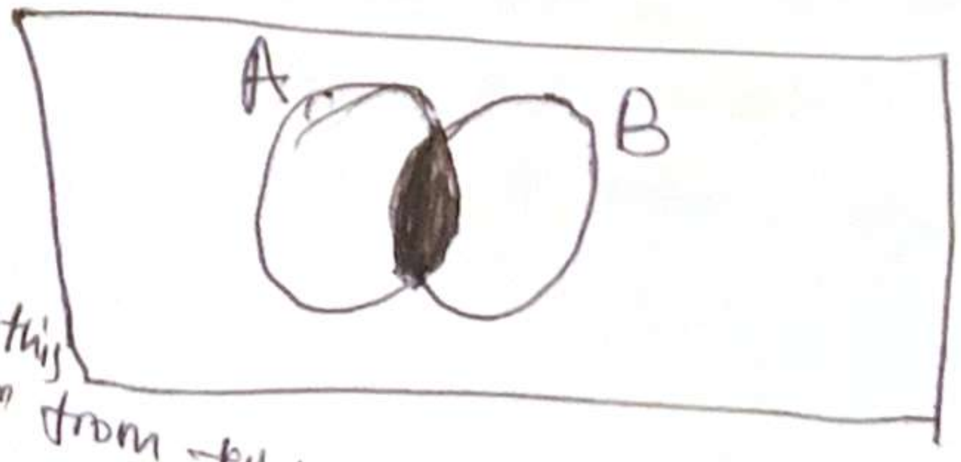
* Customer-Id is the foreign key in the Table. Because of Referred ~~key~~ Primary key.

* INNER JOIN

- Returns rows where the key exists in both tables.
- We Join two or More table on Common Column.

SYNTAX:

SELECT A. Column 1 → Taking this Column from table 1
A. Column 2
B. Column 1 → Taking this Column from table 2.
FROM table 1 AS A
INNER Join table 2 AS B
ON A. Common_Column = B. Common_Column.



→ To join Tables, There must be Common Columns.

Cont...

SQL JOINS

- Primary Key – Is a column that uniquely identify each Row.
  - No duplicates.

Customers

Customer_Id	Name
1	Rhofina
2	Lereato

Customer — Customer_Id name — customer

orders — Customer_Id amount

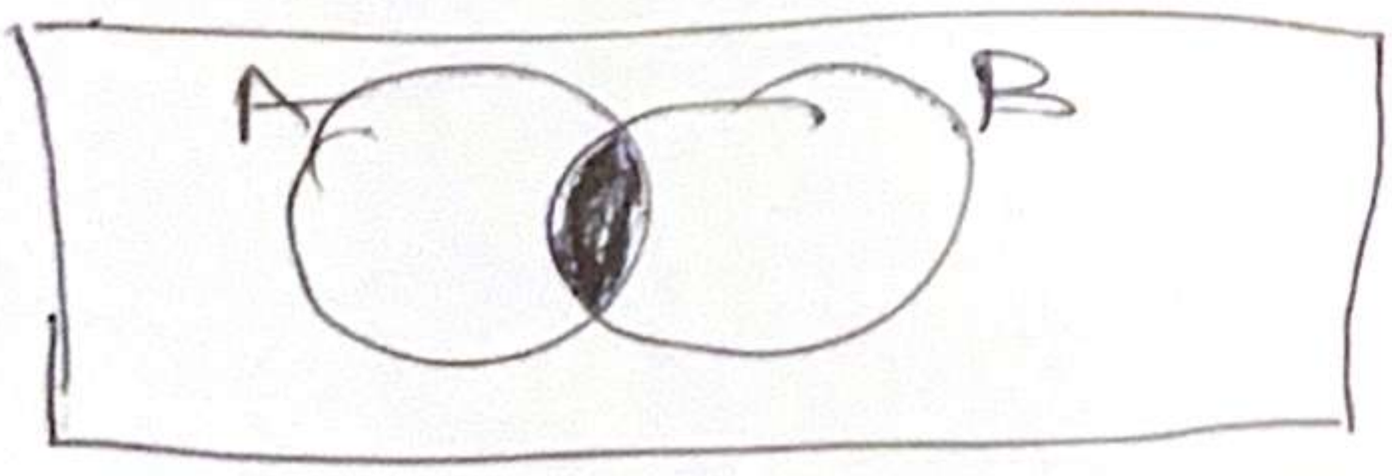
Foreign Key –

Orders → Primary key → foreign key

Order Id	Customer_Id	amount
101	1	300
102	1	150
103	3	500

INNER JOIN

- Allows us to Join two or more tables on a common Column Than gives us the Rows that are Matching in both Tables.



Whenever you Combine Tables there must be something common.

- You can Select any Columns from either of the Tables.

SYNTAX.

SELECT Column 1,
       Column 2,
       Column 3.

FROM table1 AS A
INNER JOIN table2 AS B
ON A.common_Column = B.common_Column;

Cont...

```
SELECT   A.Customer_Id,
         A.name,
         B.amount
FROM     Customers AS A — LEFT table / Base table.
LEFT JOIN Orders AS B
ON A.Customer_Id = B.Customer_Id;
```

Customer_Id	name	amount
1	Rothi	300
1	Rothi	150
2	Mbali	NULL

CORRECTION

```
SELECT
    emp_name,
    department,
    Salary,

CASE
    WHEN Salary > 80 000 THEN 'Senior IT'
        or
    WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'
    WHEN department = 'HR' AND Salary > 55 000 THEN 'Experienced HR'
    ELSE 'Staff'
END AS Position_level
FROM Employees;
```

ANID ADMINATOR SULTAN

SELECT STAMENT SYNTAX.

```
SELECT customer_name,
       amount,
CASE
   WHEN amount >1000 THEN 'High Value'
   WHEN amount BETWEEN 500 AND 999.99 THEN 'MEDIUM Value'
   WHEN amount <500 THEN 'low Value'
 END AS order_value_Category
 FROM ORDERS;
```

customer Name	Amount	order_value_Category
Alice	150	low value
BOB	560	Medium Value
Charlie	999.99	Medium value
Diana	45.50	low value
Ethan	1200.00	High Value
KOFI	NULL	NULL

NB*
1. ELSE statement is optional in THE CASE STATEMENT.
2. You put the 'ELSE' when you want default Results.
3. A statement without value is Null in SYNTAX.

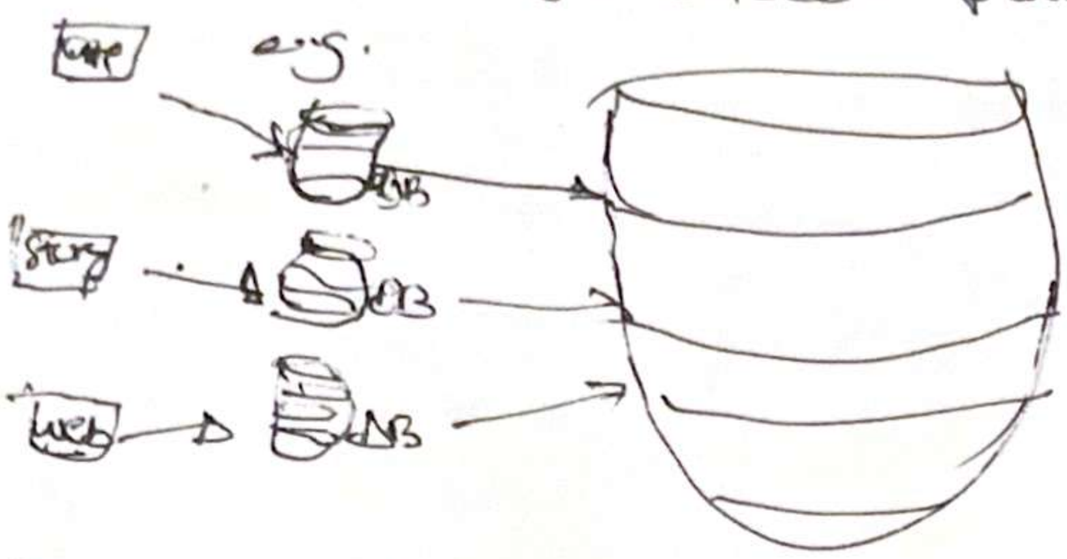
```
SELECT emp_name,
       department,
       salary
CASE
   WHEN salary > 80000 THEN 'Senior IT'
   WHEN salary >55 000 THEN 'Experienced HR'
   ELSE 'Staff'
 END AS employee_position_positional
 FROM employees
```

February 4: Data Analytics Class.

Data Collection Process.

- Whenever the Data is Collected at the back end, it Creates Unique Data ID
- ID, is a unique information that cannot be shared or accessed by someone.

- Data Base: is a Place where we Store data.
- Data warehouse is a Place where we store databases.
- Data Source: we Collect Data from Different Sources.



- Memory Card Served as a Database.
- For a memory Card to Play Songs it had to be Connected to data Manggement System.

- Data Base management Systems is what we use to access the database at the back end. eg - Sequel
- Data Base Management System is a System that we use to manage the Data within the Database.
- Data management System helps us to Control, Manage system and Read Data.
- With a Database Management System you Can manipulated data ie, Insert, delete, Update and Access.
- Example: Layaway or Database Management!
- Cashier Tills are Recording data - Serve as DataBase.
- Warehouse - Serves as a Database Management at the Back end.
- Not everyone has an access to Database at The Backend.

- The Purchasing Process - is an Online Transactional Processing. (OLTP)
- The Purchasing information is transferred to the Head Quarters Through Channel or Servers Called ETL.
- ETL Stands for: Extract, Transform, Loads.
- Online Transactional Database works during the Day.
- Online Analytical Processing Data management works at night.

AND OPERATOR SYNTAX

- SELECT Column 1,
  Column 2,
  Column 3

FROM DATABASE . SCHEMA . TABLE

WHERE CONDITION 1 AND CONDITION 2;

Curro . STUDENTS . MARKS

Student_Id	Name	Subject	Marks
101	Rhofi	Maths	99
102	Naledi	Maths	22
103	Mbali	Geo	50
104	Ally	Acc	40

QUESTION.

- We want the name and Mark of a Student doing Maths and got above 50%

SYNTAX.

SELECT Name,
Marks

FROM Curro . Student . Marks.

WHERE SUBJECT = 'MATHS' AND MARKS >50

ANSWER:

Name	Marks
Rhofi	99

OR STATEMENT

- Allows us to specify multiple conditions but only one or atleast one condition must be met.
- eg coke or Sprite, BMW or Merc, True or False.
- OR allows us to say we can take both or one.

SYNTAX.

SELECT Column 1,
Column 2,
Column 3

FROM DATABASE . SCHEMA . Table

WHERE CONDITION 1 OR CONDITION 2;

EXERCISE: CASE STATEMENT

```
SELECT Product Id, Product-name, Price,
  CASE
    WHEN Price BETWEEN 100 AND 1000 THEN 'Mid-Range'
    WHEN Price < 100 THEN 'Budget'
    ELSE 'EXPENSIVE'
    END AS Price_Category'
  FROM PRODUCTS;
```

SQL OPERATORS.

- OPERATORS, OPERATES IN THE WHERE CLAUSE.
- WE CAN USE THE OPERATORS WITHIN OUR WHERE STATEMENTS TO FILTER OUR DATA.
- FILTERING IS ABOUT INCLUDING OR EXCLUDING SPECIFIC ROWS BASED ON A SPECIFIC CONDITION.

COMPARISON OPERATORS.

OPERATOR	DESCRIPTION
=	EQUAL TO - EXACT.
!= <>	NOT EQUAL TO
> EXCLUSIVE.	>50 GREATER THAN. - ABOVE. 51, 52, 53 ...
>= INCLUSIVE.	>=50 50, 51, 53 .... GREATER THAN OR EQUAL TO, INCLUDING ANY ABOVE.
<	LESS THAN - BELOW. <50 49, 48, 47, 46 ....
=<	LESS THAN OR EQUALS TO INCLUDING AND BELOW. <=50. 50, 49, 48, 47, 46 ....

LOGICAL OPERATOR.

- THESE OPERATORS ARE MADE TO HELP US BUILD LOGIC SET OF DATA.
- AND OPERATOR: IS THE STATEMENT WE USE TO JOIN TOGETHER MULTIPLE CONDITIONS.
- e.g. FISH AND CHIPS. MATHS AND SCIENCE STUDENT.
- BOTH CONDITIONS MUST BE MET IN AN AND OPERATOR.
- SYNTAX: SELECT column1,
  column2,
  column3,
  FROM Database. Schema. Table.